Teacher Guide & Answers

Passage Reading Level: Lexile 1120

1. **C** the ability to see

2. **A** answers scientists have come up with about why some species of cave fishes are blind

3. **C** "...you have five fingers on each hand because of the genes you got from your parents. However, if you have an accident and lose a finger, your children will still be born with five fingers on each hand. If you lift weights and become a body builder, it doesn't mean your children will be born with bulging biceps. In each case, your genes haven't changed—even though your body has."

4. **B** The bigger a cave tetra's nose is, the better the cave tetra's sense of smell will be.

5. D Many cave fishes are blind, and an experiment carried out by scientists suggests that blindness in these fishes is the result of a mutation that also improves their sense of smell.

6. **A** to prepare readers for a discussion of possible answers to this question in the article

7. **D** However, it doesn't explain how a whole species of blind fish came to exist.

8. **Suggested answer**: Answers may vary but should reflect the text. The first hypothesis is that blindness gives the tetra some sort of evolutionary advantage, but not directly. That advantage might be the result of pleiotropy, which is when multiple effects are caused by the same mutation in one gene. In this case, it would mean that the gene or genes causing blindness also cause some other change that helps the fish.

9. **Suggested answer**: Another effect it might have is to improve cave tetras' sense of smell.

10. **Suggested answer**: Answers may vary but should resemble the following. Scientists' first hypothesis was that blindness gives the tetras some sort of evolutionary advantage. They thought the advantage might be the result of pleiotropy, which is when multiple effects are caused by the same mutation in one gene. The experiment led scientists to discover a gene that seems to have multiple effects. It causes blindness in cave tetras and may also improve their sense of smell. An improved sense of smell would be an evolutionary advantage for a fish that lives the dark. In this way, blindness seems to be linked to an evolutionary advantage in cave tetras. The link supports the first hypothesis scientists have about blindness in the tetras.

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